

Engineers can use modeling and simulation to make sure that yours is not just another design, but one that is optimised well. When designing a vehicle, maximising the speed and the load it can carry is a basic, rough requirement, and there are a lot of designs that will meet it. If you put an engine of a certain size or bigger you will meet the speed requirement, and if the control system of the car is set up properly the car will still remain stable in spite of the number of people in the car. However, you would not want to go overboard with the design and make it so expensive that you do not have customers. Modeling and simulation can help you ensure that your design has met all the requirements without a heavy price tag. There are so many components working behind the scenes.

What's the key to the best possible overall performance?

Modeling and simulation helps in designing a perfect combination of subsystems. For instance, when developing an electric or hybrid electric vehicle, there are a number of systems like the motor, generator, battery and engine, all of which have their own separate requirements. Simulation can help to decipher information such as how big the motor and battery should be. If the engineers want to shrink the battery by ten per cent, they can check whether they can increase the motor's size by that amount. The final design does not use the perfect motor or the perfect battery, but it is the perfect combination of the subsystems. The combination is optimised and helps the manufacturer in achieving his goals by meeting the needs of buyers and producing it as quickly and as cheaply as possible.

Where do MATLAB and Simulink come into the picture?

The person who is designing the control system of the vehicle needs to understand the entire car and has to design a control system based on specific requirements. For instance, the CAD designer wants the car to be able to accommodate a 2.5m (9-foot) tall man. So he has to build a car that is extremely tall. For the control system designer, this information is vital as a car that tall can tip over easily. He needs to find out from the mechanical department how tall the car will be and where the centre of mass will be, so that he knows what his limits are. MATLAB and Simulink are typically used in the control system as this is where everything comes together.

In most cases, you can directly import data like the centre of mass or air dynamics from other tools or, if necessary, you can use co-simulation and connect the MATLAB and Simulink environment to another tool. Control engineers

prefer to have everything under one environment so that they can optimise the design. However, co-simulation can also be used as a final verification option with as much design details as possible.

Can the CAD inputs from other software be integrated into MATLAB and modeled in a single platform?

Yes, there are many ways to get the information from CAD into MATLAB, as it can import almost any form of data. CAD systems can export a lot of elements like plain text files and lists of numbers. If you are considering three-dimensional (3D) mechanical models in the CAD system, you have the ability to export them from CAD to our tool for modeling 3D mechanical systems called SimMechanics. The CAD designer working on the static mechanical design will get the car to look as he wants while meeting the requirements of the 2.5m tall engineer, who will sit in the front seat and see that the wheels do

not smack into other parts of the suspension when the car goes up and down.

The design can be imported into SimMechanics with relevant details like mass and length, and the connections made there. You get a 3D image. Once the design is in SimMechanics, it can be connected to the electric elements, thermal elements, gear train or drive train. This

mechanical system can then be connected to the rest of the vehicle and the control system. The engineers can then test the entire design in a single environment and watch for integration challenges if any.

How are the automotive designs being turned green?

Making an energy-efficient car that produces less emission implies that the designers have to optimise their design even further. They can optimise hardware by making the frame lighter but it takes a lot of time to design and can be quite expensive to test. Software, on the other hand, is less expensive to change.

With so much pressure on making the automobiles green, automotive companies are looking for any kind of optimisation they can find. Many opportunities lie in adjusting the control system. For example, there is a very complex control system involved in running the engine. Automotive manufacturers are increasingly turning towards virtual testing and anything they can do with software, as that is where the big opportunities are. Even when they have to change the hardware, it will require a change to the control system or at least fine-tuning its parameters. That is where we hope to help them by providing the necessary tools. ●

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